2026 H2 Physics Preliminary Examination Solutions and Comments (Student)
Paper 3

1 (a) (i) Horizontally: $F\sin 40° = T\sin 24°$ ……(1)

Vertically: $F\cos 40° + T\cos 24° = 0.069g$ ……(2)

Solve: $F = 0.30632 = 0.31$ N

OR

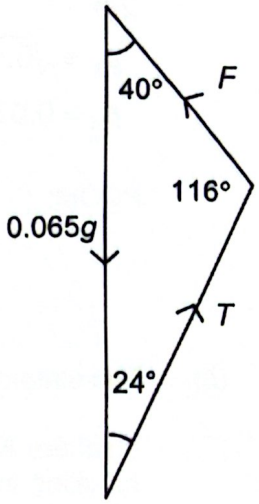
Taking moments about the point of suspension of thread supporting A,

$mg \times L\sin 24° = F\sin 64° \times L$

$$F = mg\frac{\sin 24°}{\sin 64°}$$

$$= 0.069 \times 9.81 \times \frac{\sin 24°}{\sin 64°}$$

$$= 0.306 \text{ N}$$



OR

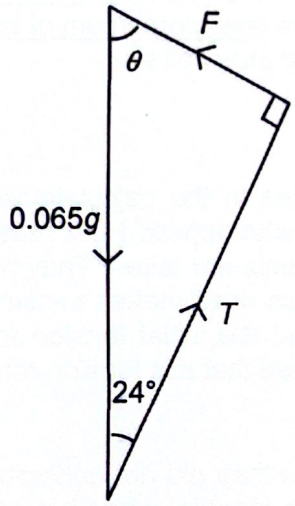
Applying sine rule to the force triangle,

$$\frac{F}{\sin 24°} = \frac{mg}{\sin 116°}$$

$$F = 0.069 \times 9.81 \times \frac{\sin 24°}{\sin 116°}$$

$$= 0.306 \text{ N}$$

(ii)



F is smallest when it is perpendicular to T.
$\theta = 90° - 24° = 66°$

(b) (i) $W = F\Delta s$ (or loss in GPE $= -mg\Delta h$)

$= (0.069)(9.81)(1.50 - 1.50\cos 24°)$

$= 0.087780 = 0.088$ J

Note: work done by gravitational force is positive as the force and the displacement are in the same direction. Also, work done by gravitational force is the same as the loss in gravitational potential energy.

(ii) Total work done = increase in KE

$$0.0878 = \frac{p^2}{2(0.069)} - 0$$

$$p = 0.11\,01$$

$$= 0.11 \text{ kg m s}^{-1}$$

(c) (i) Calculate momentum p_A of sphere A after collision.

$$\frac{p_A^{\,2}}{2m} = 52\% \times 0.0878$$

$$p_A = \sqrt{0.52 \times 0.0878 \times 2 \times 0.069)}$$

$$p_A = 0.079375 \text{ kg m s}^{-1}$$

PCOM: $\sum p_i = \sum p_f$

$$0.11 = -0.079375 + p_B$$

$$p_B = 0.189375 = 0.19 \text{ kg m s}^{-1}$$

(ii) The statement is not valid.

If sphere A loses all its initial kinetic energy, its final momentum will be zero resulting in a smaller change in momentum for sphere A as compared to when it retained some KE and rebounded.
By the principle of conservation of momentum, the change in momentum of sphere B would be smaller. Since the final momentum of sphere B is smaller, kinetic energy of sphere B is smaller after collision.

Comments

1 (a) (i) There were many careless mistakes in the calculations which led to the wrong answer of 0.29 N. Students who applied PCM must always state the point or axis about which the moments are taken. They must also state the concept applied so that the equation used makes sense to the marker, eg sine rule, etc. Many students found the initial tension to be same as the weight of sphere A and then assumed that this tension remains the same as the string is tilted.

(ii) Many students left his part blank as they did not understand that force F is the smallest (or shortest in the force triangle) when it is perpendicular to the tension T.

(b) (i) Mark is not awarded for wrong applications or statements of the principle of conservation of energy.

(ii) Students must realised to check their answers to (b)(i) if they are unable to prove 0.11 N s. No ecf can be awarded, if (b)(i) is wrong as this is a "show" question. In "show" question, student must always show the numerical value to more s.f. before rounding to the required value in the final step.

(c) (i) After taking square root, many students left the final velocity or the final momentum of sphere A as positive value (despite the question stating that it rebounds in the opposite direction). This leads to the incorrect final momentum of sphere B as 0.0307 N s.

(ii) Weaker students have no idea how to answer this part. Explanations that were vague or illegible are not awarded marks.

2 (a) Gravitational force of Earth on satellite provides centripetal force on satellite.

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$v = \sqrt{\frac{GM}{r}}$$

$$= \sqrt{\frac{\left(6.67\times10^{-11}\right)\left(6.0\times10^{24}\right)}{\left(6.4\times10^{6}+400\times10^{3}\right)}}$$

$$= 7671.6$$

$$= 7700 \text{ m s}^{-1}$$

(b)

$$U = -\frac{GMm}{r}$$

$$= -\frac{\left(6.67\times10^{-11}\right)\left(6.0\times10^{24}\right)(1200)}{\left(6.4\times10^{6}+400\times10^{3}\right)}$$

$$= -7.0624\times10^{10}$$

$$= -7.1\times10^{10} \text{ J}$$

(c) To just reach infinity, total energy of the spacecraft is always zero.

By principle of conservation of energy,

$$(\text{total energy})_{\text{at infinity}} = (\text{total energy})_{\text{at 400km}}$$

$$0 = \text{KE} + \left(-7.0624\times10^{10}\right)$$

$$\text{KE} = 7.0624\times10^{10} \text{ J}$$

$$\text{additional energy required} = 7.0624\times10^{10} - \frac{1}{2}(1200)(7671.6)^2$$

$$= 3.5311\times10^{10} = 3.5\times10^{10} \text{ J}$$

(d) (i) The kinetic energy of the spacecraft increases due to stronger gravitational acceleration at lower altitude while gravitational potential energy of the spacecraft decreases.

(ii) The total energy of the spacecraft at smaller orbital radius is more negative. Hence negative work has to be done to decrease the total energy.

Comments

2 (a) Students must show how the equation $v = \sqrt{\frac{GM}{r}}$ was derived by showing that $\frac{GMm}{r^2} = \frac{mv^2}{r}$. Substitution of values must also be shown to be awarded full credit for calculation questions. Common mistakes in this question include the omission of radius of the Earth in the value of r, using the mass of satellite as the value of M and not converting 400 km to m.

(b) Generally well done. Take note to include the negative sign as the value of gravitational potential energy should be a negative value since the satellite is bounded within the Earth's gravitational field.

(c) Generally students were able to formulate the right equation and calculation. However, the explanation of working was inadequately written as "GPE must be zero at infinity" without the consideration of kinetic energy. The initial kinetic energy that the satellite possessed at 400 km altitude should be accounted for in the calculation.

(d) **(i)** Mostly well-done, "magnitudes of both gravitational potential and kinetic energies will increase" is accepted.

(ii) Explanations that went into lengthy description of how gravitational potential (energy) is negative due to the attractive nature of gravitational force and the external force being applied in opposite direction to displacement were not accepted, without the mention of total energy becoming more negative with the decrease in orbital radius. Students must realise that while gravitational potential energy decreases with a reduced radius, kinetic energy actually increases. The total energy remains negative (despite the positive K.E.) because work is done against atmospheric resistive forces acting on the spacecraft.

3 (a) (i) Taking direction to the right of the equilibrium position as positive

By Newton's 2nd Law,

$$F_{net} = ma$$
$$k(e-x) - k(x+e) = ma$$
$$-2kx = ma$$
$$a = -\frac{2k}{m}x$$

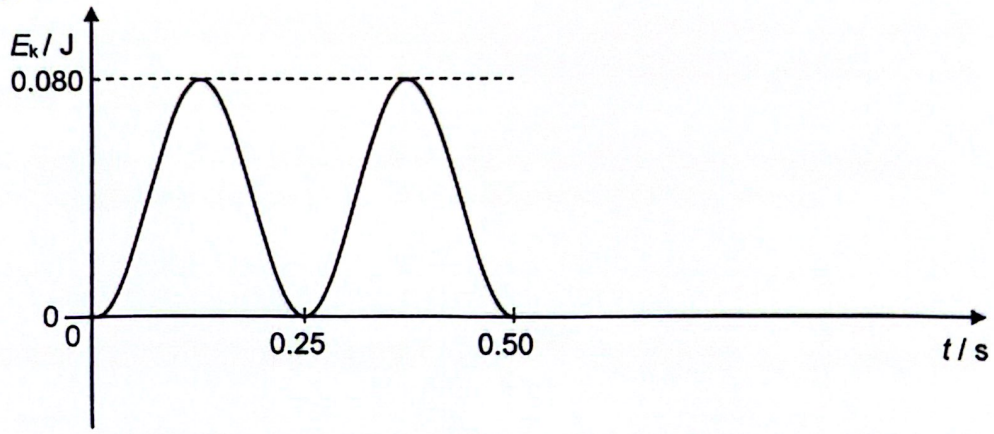
(ii) 1. For an object moving in SHM, $a = -\omega^2 x$

Since $\omega^2 = \frac{2k}{m}$,

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\frac{2k}{m}}}$$
$$= \frac{2\pi}{\sqrt{\frac{2(50)}{0.63}}}$$
$$= 0.4987 = 0.50 \text{ s}$$

2.
$$v_o = \omega x_o = \sqrt{\frac{2k}{m}}x_o$$
$$= \sqrt{\frac{2(50)}{(0.63)}}(0.040) = 0.50395 = 0.50 \text{ m s}^{-1}$$

3.



$$E_{k,max} = \frac{1}{2}mv_o^2 = \frac{1}{2}(0.63)(0.50395^2) = 0.079999 = 0.080 \text{ J}$$

Correct shape with smooth turning points + correct $E_{k,max}$ – 1M

Correct period of 0.25 s for $E - t$ graph and two complete cycles – 1M

(b) For the same applied force, the new composite spring will stretch by double the original amount. Hence, the effective force constant will be half that of the single original spring.

Consequently, the angular frequency will decrease by a factor of $\sqrt{2}$ and since the period is inversely proportional to angular frequency, the period will increase by a factor of $\sqrt{2}$.

OR

At the same displacment, the restoring force is halved as the effective force constant will be half that of the single original spring.

Consequently, the angular frequency will decrease by a factor of $\sqrt{2}$ and since the period is inversely proportional to angular frequency, the period will increase by a factor of $\sqrt{2}$.

Comments

3 (a) (i) Many students ignored the explicit condition in the question stem that the extension of each spring at equilibrium was *e*. When applying Newton's second law for a displacement *x*, students simply assumed the force on each spring was *kx*, rather than using the actual forces $k(e+x)$ and $k(e-x)$ to demonstrate how *ke* cancels out. Although a number of students managed to arrive at the given expression, no credit could be awarded as their working did not reflect a correct physical analysis of the given system.

Other common mistakes included resolving the spring forces on the trolley incorrectly and arbitrarily introducing a negative sign without physical justification simply to force their working to match the final expression to be shown.

(ii) **1.** Most students were able to determine the period successfully. The most common mistakes included assuming that $\omega^2 = \frac{k}{m}$ instead of $\omega^2 = \frac{2k}{m}$ from inspecting the derived expression in (a)(i) and forgetting to take the square root of $\frac{2k}{m}$ when calculating ω.

2. This part was mostly answered correctly. A handful of students who used energy conservation were unable to obtain the correct maximum speed because they failed to account for the new effective spring constant of $2k$ of the system given in the question. The elastic potential energy at maximum displacement should have been expressed as $\frac{1}{2}(2k)x_0^2$.

3. Common mistakes included:

- Drawing 'M'-shaped curves instead of a smooth $\cos^2$ curve starting at maximum E_k.
- Showing the wrong period for kinetic energy relative to the period *T* of the trolley's motion.

- Extending the graph into negative energy values, confusing scalar kinetic energy with sinusoidal velocity or displacement graphs.
- Drawing extra energy cycles beyond the single complete oscillation requested in the question.
- Failing to label key feature values, such as maximum E_k on the vertical axis.

(b) Students generally found this question challenging and responses varied widely. Students should begin their explanation from first principles using Hooke's law. Quoting or applying formulas for spring combinations directly without physical justification would not earn full credit.

To establish that the effective force constant of the new composite spring is halved ($k_{eff} = \frac{1}{2}k$), students needed to explain that for the same applied force, the new composite spring will stretch by double the original extension (or conversely, that a given displacement produces half the restoring force).

Derivation (not required)

A force F applied across a composite spring (a combination of 2 springs in series) acts equally on both springs. If each spring extends by $e_{single} = \frac{F}{k}$ the total extension:

$$e_{composite} = e_{single} + e_{single} = \frac{F}{k} + \frac{F}{k} = \frac{2F}{k} \Rightarrow F = \left(\frac{k}{2}\right) e_{composite}$$

Hence, the effective spring constant $k_{eff} = \frac{1}{2}k$.

Consequently, using $T \propto \frac{1}{\omega} \propto \frac{1}{\sqrt{k}}$ would allow a deduction how the period changes.

In addition, explicit numerical comparisons using the provided data are required. No credit was awarded for vague statements such as "k decreases hence T increase."

4 **(a)** **(i)** $I_1 = I_0 \cos^2\theta$

$\frac{I_1}{I_0} = 0.50$

$\cos^2\theta = 0.50$

$\theta = \cos^{-1}\sqrt{0.50} = 45°$

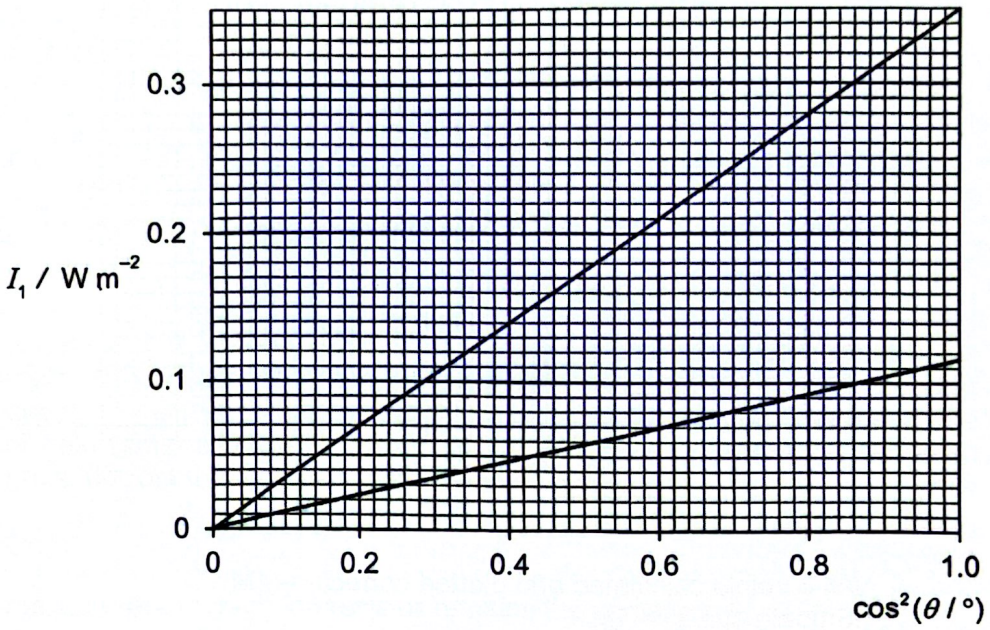
Or $180° - 45° = 135°$

Or $180° + 45° = 225°$

Or $360° - 45° = 315°$

Ans: 45°, 135°, 225° or 315° (any two)

(ii)



Gradient of I_1 with $\cos^2\theta$ gives the value of I_0
Draw straight line with gradient 1/3 of original line
(pass through origin and point (0.86, 0.100)

(b) $I \propto \frac{1}{r^2}$

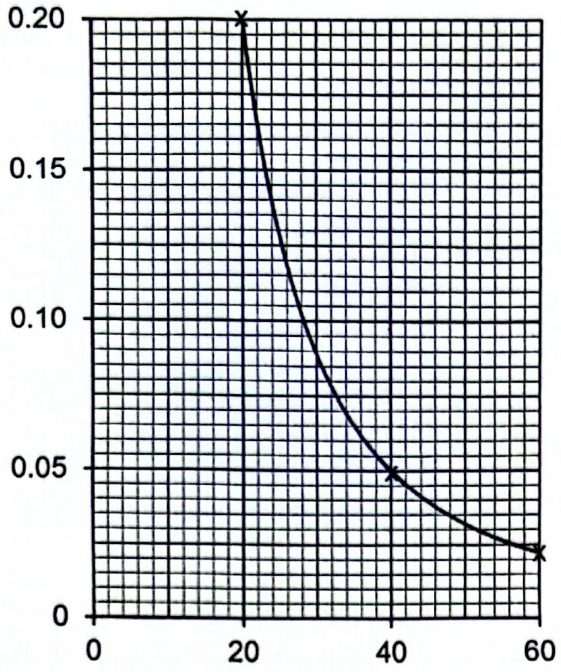
Given $I_{20} = 20$ W m^{-2}

$$I_{40} = \left(\frac{r_{20}}{r_{40}}\right)^2 I_{20}$$

$$= \left(\frac{20}{40}\right)^2 (0.20)$$

$$= 0.050 \text{ W m}^{-2}$$

$$I_{60} = \left(\frac{r_{20}}{r_{60}}\right)^2 I_{20}$$
$$= \left(\frac{20}{60}\right)^2 (0.20)$$
$$= 0.022 \text{ W m}^{-2}$$



All 3 points calculated and plotted correctly – 1M
Smooth curve – 1M
Deduct 1M for odd / inappropriate scale

Comments

4 (a) (i) Some students wrongly read off the graph and obtained the wrong value for $\cos^2\theta$ It is unnecessary to do so as the ratio of 0.50 is equal to $\cos^2\theta$.

(ii) Students should take care when calculating the new values of I_1 and draw the new line accurately. It is easier to use point(s) that fall(s) on the grid lines such as (0.86, 0.100) when drawing the new line. Do also take note to plot and read to half the smallest square.

(b) Some students did not calculate and plot accurately the values of I_d at 40 cm and 60 cm. In some cases, a small or odd scale was chosen for the intensity axis. Do label the intervals on the axis (every 10 small squares), without which the plotted points cannot be checked for accuracy.

5 (a) (i) $Q = CV = 120C\sin(314t)$

(ii) $I_C = \frac{dQ}{dt}$

$= \frac{d}{dt}(120\,C\sin(314\,t))$

$= 314(120\,C\cos(314\,t))$

$= 37700\,C\cos(314\,t)$

(b) (i) $I_{R,rms} = \frac{V_{rms}}{R} = \frac{120/\sqrt{2}}{200} = 0.424$ A

(ii) $P = \frac{1}{2}P_o \qquad (\text{or}: P = I_{rms}{}^2R)$

$= \frac{V_o^2}{2R}$

$= \frac{120^2}{2(200)}$

$= 36$ W

(c) (i) In a resistor, moving electrons continuously collide with the ionic lattice of the material. Changing the frequency changes how fast the current reverses direction, but it does not change the rate of these collisions. The mean rate of heat generated due to these collisions therefore remains the same, so r.m.s. current through the resistor remains constant.

Or:

r.m.s.current through the resistor remains the same, as $I_{R,rms} = \frac{V_o}{\sqrt{2}\,R}$ is constant/ independent of *f*.

(ii) When frequency increases, the capacitor charges and discharges much faster. Because the plates must fill up and empty out in a shorter period, the rate of flow of charge must increase. This increases the r.m.s. current through the capacitor.

Accept: r.m.s. current in the capacitor will increase, since $I_{C,rms} \propto \omega \propto f$.

Comments

5 (a) (i) Well done, a few erroneously expressed *C* instead of *Q* as subject of formula.

(ii) Well done, most correctly related I to rate of flow of charge *Q* and showed their workings fully to include the intermediate value 37680*C* cos(314*t*), which is commendable.

(b) (i) Generally well done, a few made the mistake of using 37700 or 37700*C* as peak current and obtained ridiculously huge $I_{R,rms}$ value. They did not recognise this is not the current for the resistor.

(ii) Majority gave the correct working and answer.

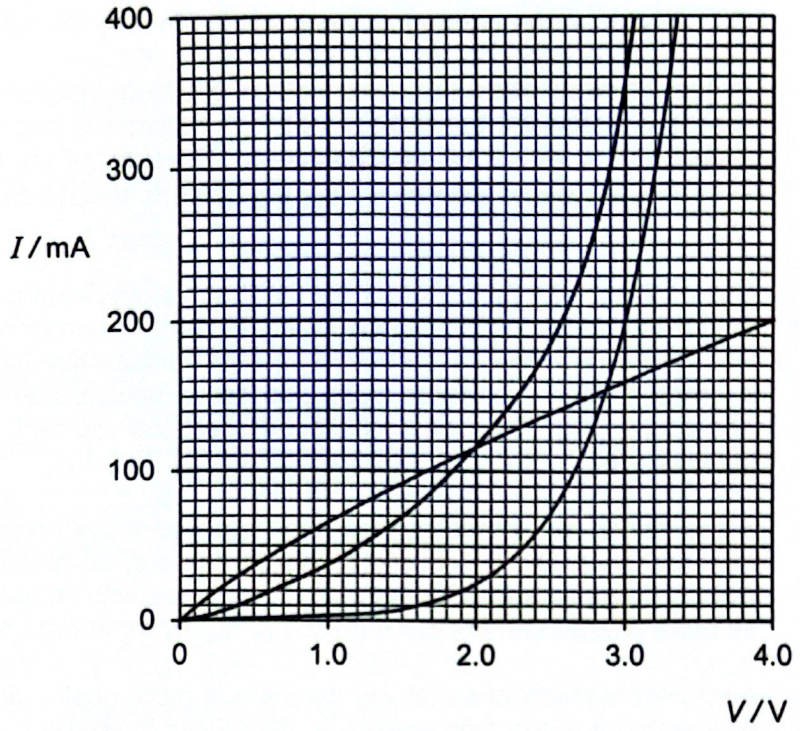
(c) (i) Most students used explanation relating to the mathematical expression for $I_{R,rms} = \frac{V_o}{\sqrt{2}\,R}$, or reasoned that since peak I_o or peak V_o is independent of f, $I_{R,rms}$ does not change. Very few used the concept of microscopic collisions between free electrons and ionic lattice to explain. Both methods are valid.

(ii) Similarly, most who successfully attempted this part used the mathematical expression for $I_{C,rms}$ and reasoned that since peak $I_{C,o}$ carries the angular frequency term ω, $I_{C,rms}$ is proportional to f and thus increases. Fewer students used the fundamental qualitative understanding that as frequency increases, the capacitor charges and discharges much faster, leading to increased rate of flow of charge.

6 (a) (i) $\text{e.m.f} = 3.00 + 4.00$
$= 7.00$ (2 d.p.)

(ii) $R = \dfrac{3.00}{0.200} = 15.0\ \Omega$ (3 s.f.)

(b) (i)



curve starting from the origin with increasing gradient and curve to be above the *I*-*V* graph for diode when $V < V_{threshold}$. –1M
From the graph, $V_{threshold} = 0.20$ V

(ii) As the current (or p.d.) increases, joule heating causes more charge carriers to be released in the thermistor but not in the resistor. Hence, the resistance of the thermistor decreases with current (or p.d.).

Comments

6 (a) (i) Students should always read the graph coordinates to the precision of half-the-smallest square. All coordinates should be written to the number of d.p. that reflects this precision.

Since the resistor and the diode are connected in series with the battery, the p.d. across the resistor and the diode should always add up to give the e.m.f. of the battery.

(ii) The resistance of an electrical component is defined as the ratio of the p.d. across it to the current that flows through it.
Resistance is not the reciprocal of the gradient of the tangent to the I-V graph as resistance is not a 'rate of change'.

(b) (i) The resistance of a negative temperature coefficient thermistor decreases with its temperature.

A thermistor is made of semiconductor material and is like a resistor where it can conduct electric current in either direction through it immediately and does not have a threshold voltage.

A diode is a made of semiconductor material with a junction that only allows electric current to flow through it in the forward-bias connection and does not allow electric current through it when it is in the reverse-bias connection.

As such, a diode has a threshold voltage due to its junction. When the diode has a p.d. below the threshold, it has high resistance and does not conduct or conducts a very small electric current. However, if the p.d. is above the threshold, the diode has low resistance and the electric current through it is large.

This explains why the I-V graph for the thermistor is above that for the diode as the thermistor conducts current immediately when connected while the diode needs a minimum (threshold) voltage before it conducts current. The I-V graph for the thermistor must also curve upwards since its resistance decreases with temperature (the larger the current, the higher its temperature).

(ii) A thermistor is a semiconductor that releases more charge carriers when temperature is high. With more charge carriers, its resistance decreases. Hence, the resistance of a thermistor decreases with increasing temperature as more charge carriers are released at higher temperatures.

A resistor is made of metal. Hence there is more obstruction to the electron flow within it when temperature is high. This means the resistance of the resistor increases with its temperature.

Due to the opposite effect on resistance with increasing temperature, the I-V graph for the thermistor is curved upwards while the I-V graph for the resistor is curved downwards.

7 (a) Inside the magnetic field, the proton experiences a magnetic force that is always perpendicular to its velocity. This force provides the centripetal force which changes the direction of velocity of the proton but the speed of the proton remains constant.

Since the magnitude of the force (dependent on speed) remains constant, the proton moves along the arc of a circle while in the uniform magnetic field.

(b) (i) The magnetic force on the proton provides the centripetal force

$$Bqv = \frac{mv^2}{r}$$

$$r = \frac{mv}{Bq}$$

$$= \frac{(1.67\times10^{-27}\times2.0\times10^{5})}{(3.5\times10^{-3})(1.60\times10^{-19})}$$

$$= 0.5964$$

$$= 0.60 \text{ m}$$

(ii) Time spent in the magnetic field

$$\Delta t = \frac{1}{4}T = \frac{\frac{1}{4}\times2\pi r}{v} = \frac{\pi(0.60)}{2\times2.0\times10^{5}} = 4.7124\times10^{-6} \text{ s}$$

$$\Delta v = \sqrt{v_i^2 + v_f^2} = \sqrt{2}v$$

$$\langle F \rangle = \frac{\Delta p}{\Delta t} = \frac{m\Delta v}{\Delta t}$$

$$= \frac{(1.67\times10^{-27})\sqrt{2}(2.0\times10^{5})}{4.7124\times10^{-6}}$$

$$= 1.002310^{-16}$$

$$= 1.0\times10^{-16} \text{ N}$$

Comments

7 (a) It is important to state that the magnetic force on the proton is **always** perpendicular to its velocity throughout its motion rather than just at the initial point of entry. Stating that the force "always points towards the same point" only describes the geometry of a center (like a radius meeting at a focal point) and does not explain the physics mechanism.

The point that the magnetic force remains constant is crucial. If the force is not constant, the radius of the path would change, causing the particle to move in a spiral rather than a uniform circle.

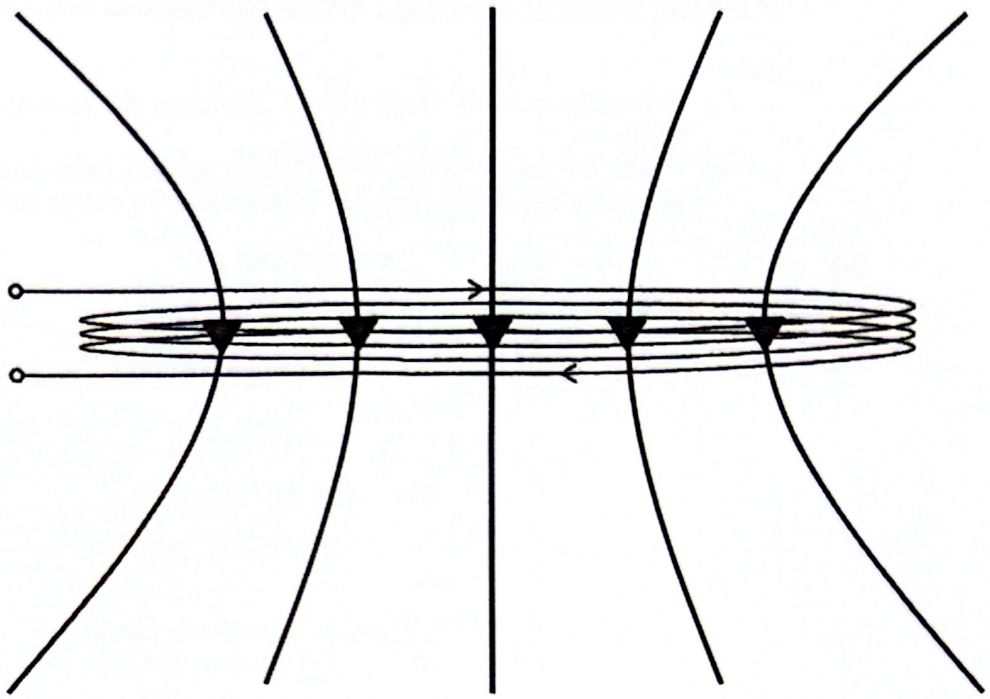
(b) (i) For 'show' questions, it is essential to show the numerical substitution and better to display the unrounded value to more s.f. than required before rounding to the final answer.

Mass of proton is not the same as unified atomic mass.

(ii) The average force in this question is not the magnetic force / centripetal force. It is $\frac{m\Delta v}{\Delta t}$ where Δt is the time taken to travel ¼ of a circle.

8 (a) Magnetic flux density is the force acting per unit current per unit length on a conductor placed perpendicular to a magnetic field.

(b) (i)



(deduct 1M for every mistake)

- Correct direction (pointing downwards)
- At least 5 lines - Straight line in the centre, other 4 lines to spread outwards, different radius. The further the line from the centre, the smaller the radius of the arc
- No more than 2 parallel lines in the centre
- Pattern must be symmetrical about the centre
- Curvature should be symmetrical above and below the coil
- Field lines in the coil should be equally spaced
- Field lines must be continuous and not broken
- Lines must occupy the entire coil
- Field lines should not converge or look like they will converge

(ii) 1. The leftward current in the straight wire is perpendicular to the magnetic flux density of the coil, which is directed into the plane of the paper. By Flemming's Left Hand Rule, the wire experiences a magnetic force vertically downwards perpendicular to both the current and the magnetic field.

Hence, the wire moves downwards.

Note:
Direction of force on wire must be correct in order for explanation to be valid.

2. There is no effect on the wire as the direction of the magnetic field of the coil along the axis is parallel to the direction of the current in the wire.

(c) (i) The high frequency alternating current sets up a high frequency oscillating magnetic field in the flat circular coil. As a result, the magnetic flux through the base of the saucepan changes at a high frequency.

According to Faraday's law, a large emf is induced in the base of the saucepan. Since the base of the pan is a conductor with resistance, a large eddy current is produced. This gives rise to Joule heating and a large amount of thermal energy is dissipated, heating the contents in the saucepan.

Note:

- *Reference must be made to high frequency AC leading to large currents produced.*
- *Reject answers that use $P = V^2/R$ to explain large power dissipation in saucepan without any mention of Joule heating due to induced currents.*

(ii) 1.

$$(I_{induced})_{max} = \frac{|E_{max}|}{R}$$

$$= \frac{1}{R}\left|-\frac{d\phi}{dt}\right|_{max}$$

$$= \frac{1}{R}\left(\frac{dBA}{dt}\right)_{max}$$

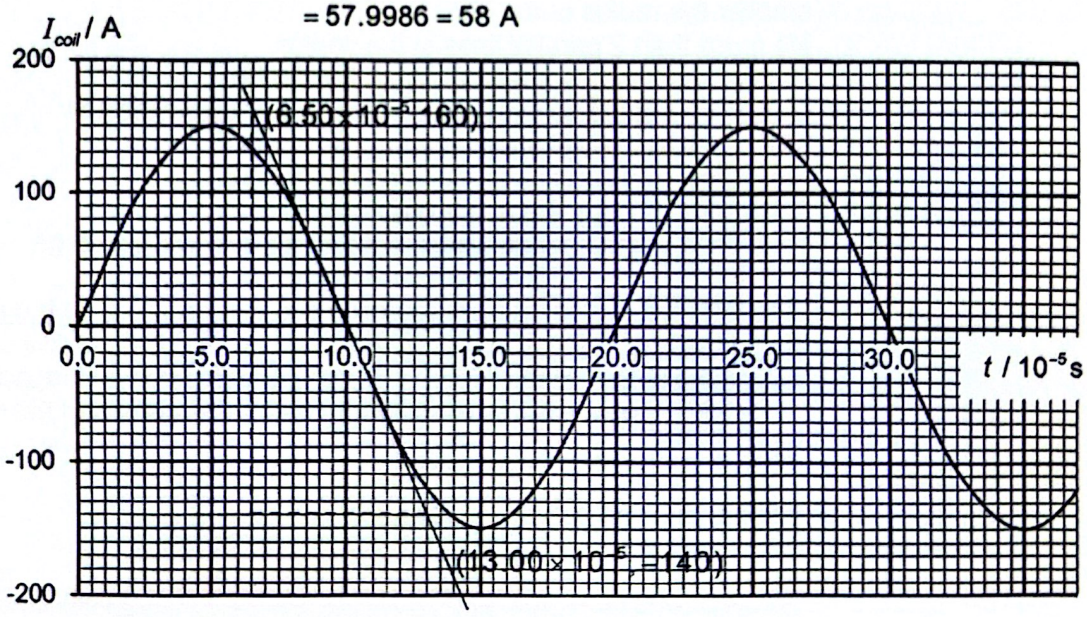
$$= \frac{A}{R}\left(\frac{dB}{dt}\right)_{max}$$

$$= \frac{A}{R}\left(\frac{d(5\mu_o N I_{coil})}{dt}\right)_{max}$$

$$= \frac{5\mu_o NA}{R}\left(\frac{dI_{coil}}{dt}\right)_{max}$$

$$= \frac{5(4\pi\times10^{-7})(20)(0.020)}{(0.20)}\left|\left(\frac{-140-160}{13.00\times10^{-5}-6.50\times10^{-5}}\right)\right|$$

$$= 57.9986 = 58 \text{ A}$$

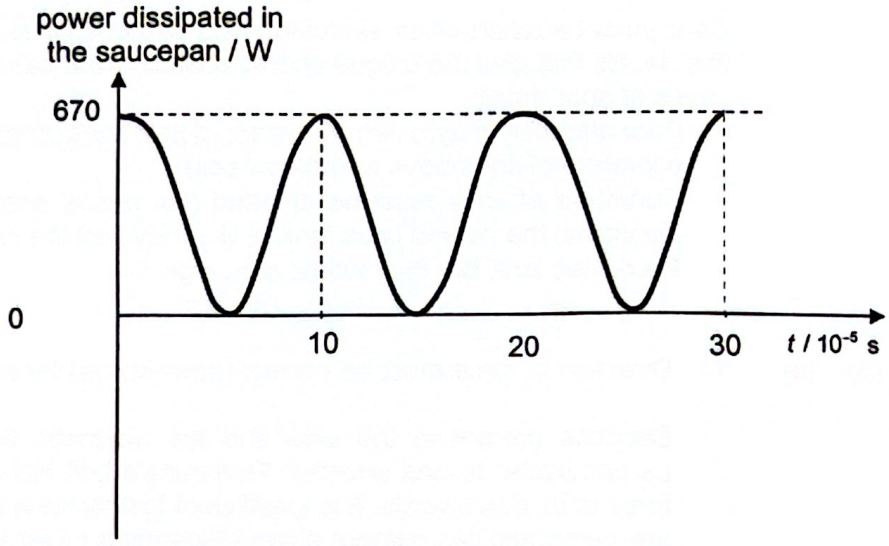


OR

$$(I_{induced})_{max} = \frac{|E_{max}|}{R}$$
$$= \frac{1}{R}|B_{max}A\omega|$$
$$= \frac{1}{R}(5\mu_o N I_{coil,max})(A)\left(\frac{2\pi}{T}\right)$$
$$= \frac{(5)(4\pi\times10^{-7})(20)(150)(0.020)(2\pi)}{(0.20)(20\times10^{-5})}$$
$$= 59.23 = 59 \text{ A}$$

- *Acceptable range: 55 – 63 A (Tolerance of approx ±5%)*
- *Coordinates should be read off to correct dp.*

2. $P_o = (I_{induced,max}{}^2 R)$
$= (57.9986^2)(0.20)$
$= 672.77 = 670$ W



Cosine squared graph – 1M
Correct period (3 cycles shown) – 1M
Correct value of peak power (ecf from (b)(ii)1.) – 1M

3. $\langle P \rangle = \frac{1}{2}P_o$
$Q = \langle P \rangle t = mc\Delta T$
$\Delta T = \frac{\langle P \rangle t}{mc}$
$= \frac{0.50\times672.77\times(4.0\times60)}{0.900\times2000}$
$= 44.851 = 45$ °C

(iii) The high magnetic permeability of stainless steel maximises the (rate of change of) magnetic flux linkage between the flat circular coil and base of the cookware. This results in large induced emf / eddy currents induced in the cookware.

- *Accept: increases the rate of change of magnetic flux linkage*
- *Insufficient to say increases magnetic flux density in the cookware without linking it to increase magnetic flux linkage.*

Comments

8 (a) All underlined phrases must be stated to gain credit.

It is incorrect to say 'force per unit length on a conductor per unit current'. This changes the meaning and also units of magnetic flux density. It must be strictly 'per unit length per unit current' and not separated.

(b) (i) Generally not well drawn.

Take note of all the marking points. These are referenced from the examiners report.

Care must be taken when sketching field patterns because there are many fine details that give the unique characteristics of the pattern. Some common cases of poor detail:

- Poor attention to symmetry (number of field lines on either side of pattern, curvature of line above and below coil)
- Curvature of lines must be different (i.e. radius gets smaller). In many diagrams, the curved lines looked like they had the same radius.
- Field lines look like they would converge

(b) (ii) 1. Direction of force must be correct (downwards) for any credit.

Because current in the wire and the magnetic field of the coil are perpendicular to one another, Flemming's Left Hand Rule predicts the force to be downwards. It is insufficient to only state the current and field are perpendicular, without stating Flemming's Left Hand Rule and vice versa. Both points must be included.

It is incorrect to explain the direction of the force by talking about current in the top of the coil being opposite in direction to that in the wire, hence leading to 'repulsion'. The correct concept should be about the interaction between the current of the wire and field of the coil.

2. Generally well done.

(c) (i) Many answers mentioned that a large emf is induced in the base of the saucepan, and used $P = V^2/R$ to explain that this meant a large power dissipated.
While this is mathematically correct, credit was not awarded. The physical mechanism for energy transfer to the saucepan is the induced current, through the process of Joule heating. Emf itself does not guarantee heating. Only when current flows will there be heating.

(ii) **1.** The tedious calculation led to many careless mistakes.
Students who used the maximum current in the coil to calculate the maximum induced current tended to make less careless mistakes than those who used the gradient of the graph.

2. Some students still cannot sketch a cosine squared graph, drawing instead the modulus cosine graph.

3. Some students did not use average power.

(c) (iii) It is insufficient at the A Level to say that the field lines are 'concentrated'.

9 **(a)** A stationary wave is formed by the superposition of two progressive waves of the same type, amplitude, frequency and speed, travelling along the same path but in opposite directions.

(b) **(i)** **1.** $\text{Amplitude} = \frac{7.0}{2} = 3.5 \text{ cm}$

2. $\frac{T}{2} = 9.0 \times 10^{-3} \text{ s} \Rightarrow T = 18.0 \times 10^{-3} \text{ s}$

$f = \frac{1}{T} = \frac{1}{18.0 \times 10^{-3}} = 55.56 = 56 \text{ Hz}$

3. $v = \frac{\lambda}{T} = \frac{0.240}{18.0 \times 10^{-3}}$

$= 13.33 = 13.3 \text{ m s}^{-1}$

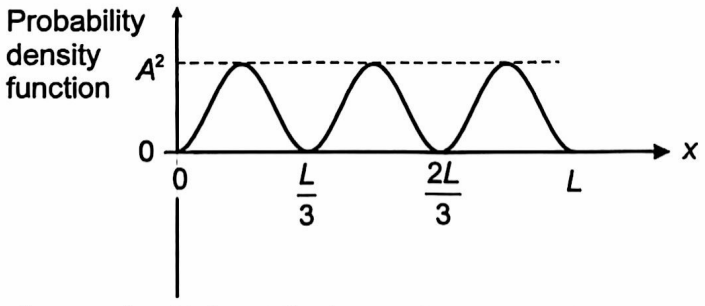
(ii) **1.** 0 rad
cannot accept no phase difference since answer line has units rad.

2. 3.14 rad (accept π rad)

(c) **(i)** $\int_0^L |\psi|^2 dx$ is the probability of finding the particle between $x = 0$ and $x = L$ of the box. As the box is perfectly reflecting, and the particle is trapped within the box, the probability of finding the particle within the box has to be 1. Hence $\int_0^L |\psi|^2 dx = 1$.

(ii) $\lambda_n = \frac{2L}{n}$

(iii)



$\sin^2$ curve (no kinks at the base of the humps) for 3 complete cycles – 1M
Appropriate labels L/3, 2L/3 and L and A^2 – 1M

(iv)

$$\int_0^L |\psi_n(x)|^2 dx = \int_0^L A^2 \sin^2 \frac{2\pi x}{\lambda_n} dx$$

$$= \frac{A^2}{2}\int_0^L 2\sin^2 \frac{2\pi x}{\lambda_n} dx$$

$$= \frac{A^2}{2}\int_0^L (1-\cos\frac{4\pi x}{\lambda_n}) dx$$

$$= \frac{A^2}{2}\left[x - \frac{\lambda_n}{4\pi}\sin(\frac{4\pi x}{\lambda_n})\right]_0^L$$

$$= \frac{A^2}{2}\left[x - \frac{2L}{4n\pi}\sin(\frac{4n\pi x}{2L})\right]_0^L$$

$$= \frac{A^2}{2}[L+0] = 1$$

$$\therefore A = \left(\frac{2}{L}\right)^{\frac{1}{2}}$$

OR

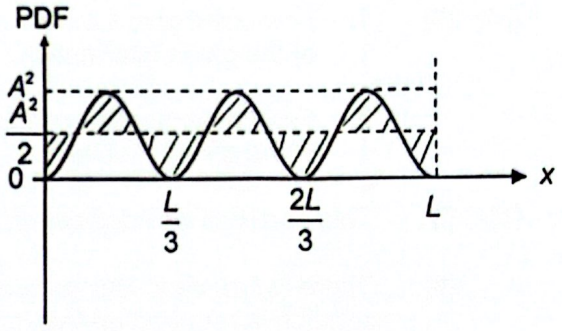
Using graphical method

From (c)(iii)

Since $\int_0^L |\psi_n(x)|^2 dx = 1$

$$\frac{A^2}{2} \times L = 1$$

$$A = \sqrt{\frac{2}{L}}$$



(v) If the walls were not perfectly reflecting, it means that the wave function ψ is non-zero at the walls ($x = 0$, $x = L$). Thus, the particle can penetrate the potential barrier walls and the wave function will decay exponentially within the wall (potential barrier).

Hence there is a finite probability that the particle can be found beyond the boundaries of the box.

(d) (i) The electrons arrive as single (discrete) particles as shown clearly in the first picture.
OR
The dots show that the electrons arrive as single (discrete) particles.

At longer exposure (20 min, 51 s, 5 min 42 s), the regular maxima and minima characteristic of an interference pattern emerges, showing that they exhibit wave-like behaviours.

(ii) Regular fringe separation implies constant wavelength.

According to the de Broglie's postulate, the wavelength of a particle is related to its momentum by $\lambda = \frac{h}{p}$ where h is the Planck's constant. Since wavelength is constant, the momentum of the electrons is constant.

which in turn implies that the electrons have the same velocity.

Comments

9 (a) Most students could not write out the full correct answer, there were quite a lot of incomplete answers.

(b) (i) 1. Few students managed to obtain the correct answers, thus showing a lack of understanding of the given information.

2. Few students managed to obtain the correct answers, thus showing a lack of understanding of the given information.

3. Few students managed to obtain the correct answers, thus showing a lack of understanding of the given information.

(b) (ii) 1. Few could give the right answers, again showing a lack of understanding of the given information.

2. Few could give the right answers, again showing a lack of understanding of the given information.

(c) (i) This part was well answered.

(ii) There were very few correct answers to this part of the question. More revision is required on this topic.

(iii) There were very few correct answers seen. More revision is needed for this topic.

(iv) Very, very few students showed the correct working for this part. The majority of students failed to display a clear working leading to their final answer.

(v) The majority of students could not answer this part correctly. More revision is needed on this topic.

(d) (i) This part was mostly well done.

(ii) Very few students answered this part well. A significant number of students incorrectly introduced Heisenberg's Uncertainty Principle into their responses, leading to confused arguments. Students must clearly distinguish between the de Broglie relation for matter waves and quantum uncertainty principles. It was evident that much confusion exists surrounding these core concepts.